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Grade	Not yet graded

Question 1

Correct

Mark 1.00 out of 1.00

Indicate which of the following statements is NOT an objectives of a modern and secure cryptographic system:

- ☐ a. Ensures data integrity
- ☐ b. Authenticates communication parties
- ☐ c. Non-repudiation of messages
- ☐ d. Ensure data integrity
- ☒ e. Implement cryptographic algorithms developed in secret by teams of specialists ✓
- ☐ f. Ensures total or partial confidentiality

The correct answer is: Implement cryptographic algorithms developed in secret by teams of specialists

Question 2

Incorrect

Mark 0.00 out of 1.00

You are receiving an important secret message ([see attachment](#)) which is encrypted using AES (ECB mode, PKCS7 padding).
You already have the key: 12345678password

Decrypt the message and select the correct (exactly) content

- ☒ a. This is a secret message. ✗
- ☐ b. This is a Secret Message.
- ☐ c. THIS IS A SECRET MESSAGE.
- ☐ d. This is a SECRET message.

Your answer is incorrect.

The correct answer is:

This is a SECRET message.

Question 3

Correct

Mark 1.00 out of 1.00

What is the decimal value of 00010101

- ☐ a. 23
- ☐ b. 19
- ☒ c. 21 ✓
- ☐ d. 22
- ☐ e. 20

Your answer is correct.

The correct answer is:

21

Question 4

Correct

Mark 1.00 out of 1.00

This is a mode of operation for a block cipher, with the characteristic that each possible block of plaintext has a defined corresponding ciphertext value and vice versa.

- ☐ a. footprinting
- ☒ b. Electronic Code Book ✓
- ☐ c. Cipher Block Chaining
- ☐ d. watermark
- ☐ e. hash function

The correct answer is: Electronic Code Book

Question 5

Complete

Marked out of 1.00

What is the security strength (in bits) of a password which has the next characteristics

- has a size of 10 characters
- uses digits from 0 to 9
- uses small cap letters
- uses only !

Explain how you computed the value

First we count how many different inputs we can have. 0-9 means 10 + small letters = 26 + 1 from ! = 37, then we apply the formula with $\log_2$ that I don't remember.

Question 6

Correct

Mark 1.00 out of 1.00

Asymmetric key cryptography is used for all of the following except:

- ☐ a. Authentication
- ☐ b. Encryption of data
- ☒ c. Steganography ✓
- ☐ d. Nonrepudiation
- ☐ e. Access control

The correct answer is: Steganography

Question 7

Incorrect

Mark 0.00 out of 1.00

What is the size of the SHA-1 hash in bytes ?

- ☒ a. 16 ✖
- ☐ b. 20
- ☐ c. 32
- ☐ d. 24
- ☐ e. 28

Your answer is incorrect.

The correct answer is:

20

Question 8

Correct

Mark 1.00 out of 1.00

Which of the following components decides how secure an encryption protocol is?

- ☐ a. The number of keys used by the encryption protocol
- ☐ b. The number of keys used by the algorithm
- ☐ c. Number of users in your system
- ☐ d. None of the other answers
- ☒ e. Number of bits used for encryption by the algorithm ✔

The correct answer is: Number of bits used for encryption by the algorithm

Question 9

Correct

Mark 1.00 out of 1.00

Vigenere encryption system is a

- ☐ a. a mechanical encryption system
- ☒ b. poly-alphabetic substitution cipher ✔
- ☐ c. none of these options
- ☐ d. mono-alphabetic substitution cipher
- ☐ e. a public key cipher

The correct answer is: poly-alphabetic substitution cipher

Question 10

Incorrect

Mark 0.00 out of 1.00

You are a British soldier in the 1st World War. You receive an encrypted message using the Playfair algorithm. Knowing that the password for today is KING, decrypt GUUGBIGUMGDF. You are using a 5x5 matrix with default settings.

What are the orders?

- ☐ a. don't attack
- ☒ b. the decrypted message makes no sense in English ❌
- ☐ c. retreat
- ☐ d. attack at once

Your answer is incorrect.

The correct answer is:

attack at once

Question 11

Incorrect

Mark 0.00 out of 1.00

Which of the following is true?

- ☐ a. You need just one single key for encrypting as well as decrypting in PK cryptography
- ☐ b. None of the other answers
- ☐ c. PK cryptography is a very complex and nearly impossible setup
- ☒ d. Public key can only encrypt and private key can only decrypt ❌

The correct answer is: None of the other answers

Question 12

Correct

Mark 1.00 out of 1.00

You are a U-boat commander and you want to transmit a message to the base.

You are using an M3 Enigma machine. The key of the encrypted message is

- AGD for the initial rotor settings
- additional D->F and H -> M direct plugs
- the other settings are default

If you want to send HELP what you will be the encrypted message?

- ☐ a. ZKSS
- ☒ b. NKSS ✓
- ☐ c. LZDF
- ☐ d. LZFD

Your answer is correct.

The correct answer is:

NKSS

Question 13

Correct

Mark 1.00 out of 1.00

You want to generate a HMAC for an encrypted file ([check attachment](#)) that you want to send to your organization

The message is available in the attachment

What will be the HMAC value for it, considering

- SHA-256 for the hash function
- secretpassword for the key

- ☐ a. C4 45 8F 63 11 A3 7E 90 38 24 52 6F C5 B9 3A 39 7A BA BB 4F B2 DB 74 CD 9B 01 78 B6 14 8C 69 E0
- ☒ b. 44 52 9C AC 0C 0E 7D E7 77 F7 24 73 56 13 9E 8A 8A DE CC C6 40 54 BA E3 03 42 85 0E AC 44 B4 97 ✓
- ☐ c. CF A3 44 B3 E4 A3 F0 F1 21 78 B9 62 6B 9A 5E 65
- ☐ d. 98 4E 00 FE F8 DF 05 6D 8C 3A 03 77 6C FB 56 36 B7 BF C4 A7 8D 5F 7D 75 91 9F 0F 0D C3 AA 6A 89
- ☐ e. 60 91 E3 CB EA E7 6A 70 69 E1 F7 6E 29 27 B1 12 AF BC 39 A0

Your answer is correct.

The correct answer is:

44 52 9C AC 0C 0E 7D E7 77 F7 24 73 56 13 9E 8A 8A DE CC C6 40 54 BA E3 03 42 85 0E AC 44 B4 97

Question 14

Correct

Mark 1.00 out of 1.00

You received a secret message ([check attachment](#)) and you want to check if some one has been able to change it. You already have the SHA-1 value for it. Which one is it ?

- ☐ a. B2 88 A0 1D EF 28 B5 69 26 D0 F3 44 02 40 E3 6C E8 86 84 3A
- ☐ b. 44 64 A5 90 A9 B7 18 4D 74 A8 AF C2 83 A5 17 F6 E8 A5 11 E0 09 F8 85 0B 01 23 B5 29 D9 18 5E 58
- ☐ c. 03 B9 63 C2 24 0F 4C 4C 02 F0 E2 61 01 EA 41 32
- ☐ d. A5 77 AF 15 AA 28 B5 69 26 D0 F3 44 02 40 E3 6C E8 86 84 3A
- ☒ e. A2 88 AE 10 EE 28 B5 69 26 D0 F3 44 02 40 E3 6C E8 86 84 3A ✓

Your answer is correct.

The correct answer is:

A2 88 AE 10 EE 28 B5 69 26 D0 F3 44 02 40 E3 6C E8 86 84 3A

Question 15

Correct

Mark 1.00 out of 1.00

A Hash value is used for:

- ☐ a. To guarantee the message source
- ☐ b. To prevent situations in which the source does not recognize its message
- ☐ c. To ensure the identity of of the parties
- ☐ d. To hide the message content
- ☒ e. To guarantee message integrity ✓

The correct answer is: To guarantee message integrity